

Visal Gunawardena

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PROFILE

A dedicated and enthusiastic individual studying at the University of Moratuwa, passionate about mechanical systems and aerodynamics. I have successfully performed as a leader and as an enthusiastic, curious team member with a positive communication style. I am confident in my ability to adapt to any condition with a never-giving-up, fighting mindset. Alongside my academic interests, I am skilled in drawing and actively involved as a sportsman, maintaining discipline, teamwork, and resilience both on and off the field.

EDUCATION

University of Moratuwa	2022 - present
Bsc. Mechanical Engineering (Hons.)	GPA: 3.34/4.0
Kingswood College - Kandy	2019 - 2021
Passed G.C.E Advanced Level obtaining 3 A's	Z Score: 2.1107
Wesswood International College - Kandy	2008 - 2018
Passed G.C.E Ordinary Level obtaining 8 A's and 1 B.	

PROJECTS

Sliding Mesh Gearbox design for a Corn Shredder, (Ongoing Semester 5 Project)

Currently designing a sliding mesh gearbox with three forward and one reverse gears for a corn shredding machine, addressing challenges identified in Monaragala's corn cultivating areas. The project follows a structured design methodology including problem analysis, concept development, gear calculations, and 3D modeling in SolidEdge and SolidWorks. We are also exploring key considerations such as manufacturability, safety, lubrication, and assembly planning as part of the ongoing development process.

Production and Process Analysis of Perez Leather Artisans, (Ongoing Semester 5 Project)

Currently analyzing small leather goods manufacturing operations to identify delays, unused resources, and process problems. This involves collecting and reviewing production, financial, and workforce data to find ways to improve efficiency, reduce costs, and minimize material waste. Tecnomatix Plant Simulation is being used to model the production process and suggest continuous improvements to boost productivity, quality, and sustainability.

GridX Stem Challenge 2025

Designed and built a CO_2 powered mini race car using Fusion 360 for modeling and ANSYS for aerodynamic analysis, and secured Runner-up position in the Fastest Car category.

IMechE Railway challenge 2026, (Ongoing competition)

Participating in the IMechE Railway Challenge 2026, currently laying the groundwork and initial development for the project. Engaged in conceptual design, planning, and preparatory tasks as part of the ongoing development process.

Reverse Engineering of a Table Lamp, (Semester 4 Project)

Developed a SolidWorks prototype of a table lamp to analyze its manufacturing processes, performed material testing and stress analysis to evaluate performance and durability, and proposed design modifications to enhance functionality, aesthetics, and manufacturability.

Filerover: Automated files distributing robot, (Semester 2 Mechatronic Project)

Developed a fully automated file-distribution robot capable of collecting and delivering documents within an office environment, while monitoring the availability of responsible personnel to ensure timely and accurate delivery.

Antigravity table, (Semester 2 Manufacturing Project)

Designed and fabricated an antigravity table using woodworking and welding, integrating multiple manufacturing techniques and assembly processes to produce a durable and aesthetically pleasing product. Gained hands-on experience in precision, craftsmanship, and teamwork throughout the manufacturing process.

SOCIETIES AND LEADERSHIP

Batch Representative of 22nd Batch, Department of Mechanical Engineering 2025 - Present

- **Mechanized 24'** – Organizing Lead
- **Sililara Handawa** - Organizing Committee

University Basketball Team, Member 2022 -Present

- Part of the winning team at Sri Lanka University Games (SLUG), 2025
- **LTL 5x5 Challenge Trophy** – Organizing Committee
- **MyCola 3x3 Challenge Trophy** - Organizing Committee

IESL Student Chapter, University of Moratuwa 2024

- **Exhibition Committee** – Pillar Member
- **Compere**

Batch Representative (E group), Engineering Faculty 2022

- **Faculty Pirith Ceremony** – Organizing Lead

SKILLS

Software Applications

- AutoCAD, Solid Works, Solid Edge, MATLAB, ANSYS, Arduino, Fusion 360

Programming Languages and Frameworks

- Python, C++

Languages

- Fluent: English, Sinhala
- Basic: Tamil

References

Dr. Lihil U. Subasinghe

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

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